

20. Attributes aur Properties

Do alag cheezein

- **Attribute** = jo **HTML** me **likha** hai.
- **Property** = jo **DOM object** me hai.

Browser HTML parse karke DOM banata hai, aur **standard** attributes ke liye automatically properties bana deta hai (`<body id="page">` -> `body.id === "page"`). Par ye mapping hamesha ek jaisi nahi hoti.

DOM Properties

DOM nodes normal JS objects hain, apni property/method jod sakte ho:

js

```
document.body.myData = { name: 'Caesar' };  
Element.prototype.sayHi = function() { alert(this.tagName); };
```

Ye kuch bhi value ho sakti hain aur **case-sensitive** hain.

HTML Attributes

- **Non-standard** attribute ke liye property nahi banti:

html

```
<body id="test" something="non-standard">
```

`document.body.something` -> `undefined`.

Attributes ke saath kaam karne ke methods:

- `elem.hasAttribute(name)`
- `elem.getAttribute(name)`
- `elem.setAttribute(name, value)`
- `elem.removeAttribute(name)`
- `elem.attributes`: sab attributes ka iterable collection

Attributes ki khasiyat:

- Naam **case-insensitive** (`id` = `ID`)
- Value hamesha **string** (`setAttribute('Test', 123)` -> `"123"`)
- Sab `outerHTML` me dikhte hain

Property aur Attribute ka sync

Zyaadatar dono taraf sync hota hai:

```
js
input.setAttribute('id', 'id');    // input.id = 'id'
input.id = 'newId';                // getAttribute('id') = 'newId'
```

Exception: `input.value` sirf attribute -> property sync hota hai, ulta nahi:

```
js
input.setAttribute('value', 'text'); // input.value = 'text'
input.value = 'newValue';            // attribute 'text' hi rehta hai
```

Ye useful hai: user badal de to bhi HTML ki original value attribute me bachi rehti hai.

Properties typed hoti hain

Zyaadatar string, par kabhi kabhi nahi:

- `checkbox.checked`: **boolean** (attribute wahi empty string)
- `style` attribute = string, `style` property = **object** (`div.style.color`)
- `href` property hamesha **poora URL** deti hai, attribute wahi jo HTML me likha (`#hello`). "Jaisa likha waisa" chahiye to `getAttribute('href')`.

Custom data: `data-*` aur `dataset`

Non-standard attribute future me standard ban sakta hai (conflict ka risk). Isliye `data-` se shuru hone wale attributes programmers ke liye reserved hain:

```
html
<div id="order" data-order-state="new"></div>
```

```
js
order.dataset.orderState;           // "new" (camelCase)
order.dataset.orderState = "pending"; // badalna bhi allowed
```

CSS me bhi use kar sakte ho: `.order[data-order-state="new"] { color: green; }`

Summary

	Properties	Attributes
Type	Kuch bhi (standard ke typed)	Sirf string
Naam	Case-sensitive	Case-insensitive

- Zyaadatar cases me **properties** use karo.
- **Attributes** tab use karo jab: non-standard attribute chahiye ((`data-*`) ho to (`dataset`)), ya HTML me likhi original value chahiye.

Tasks se

- `data-widget-name` padhna: `elem.dataset.widgetName`
- External links dhundhna: `a[href*="://"]:not([href^="http://internal.com"])`. Yaha `getAttribute('href')` use hota hai (property nahi) kyunki raw value chahiye.